Blinded by Generated Contexts: How Language Models Merge Generated and Retrieved Contexts When Knowledge Conflicts?

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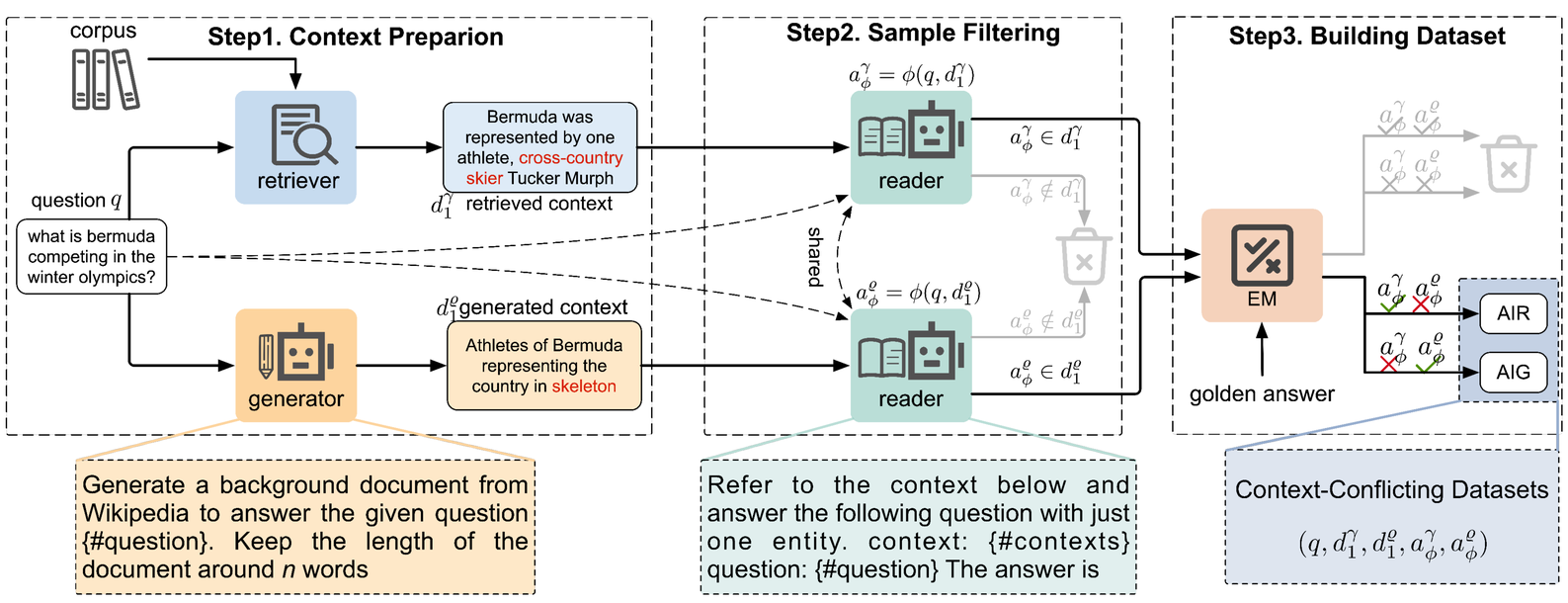
ACL 2024 Bangkok, Thailand

LLM are biased toward LLM-generated contexts even when they are wrong.

For the full paper
Just scan!

Context-Conflicting Datasets (CC)



Only one of generated and retrieved contexts contain correct answer.

- **AIG**: **A**nswer **I**n **G**enerated Contexts
- **AIR**: **A**nswer **I**n **R**etrieved Contexts

DiffGR the extent of LLMs' tendency to rely on generated contexts over retrieved contexts.

$$\text{DiffGR} = \frac{\rho_{\text{gen}} - \rho_{\text{ret}}}{\rho_{\text{gen}} + \rho_{\text{ret}}}$$

- **Ideal** DiffGR·**1 → AIR, 1 → AIG**

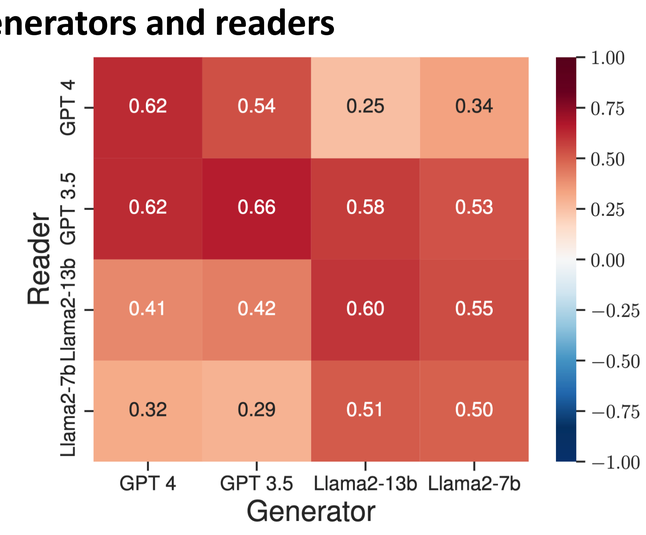
LLMs Prefer Generated Contexts

1. LLMs prefer generated contexts, even when they are incorrect!

EM performance drops significantly in AIR.

Generator &Reader	NQ-CC		TQA-CC	
	NQ-AIG	**NQ-AIR**	TQA-AIG	**TQA-AIR**
GPT-4	91.34	**17.69**	94.57	**19.09**
GPT-3.5	91.85	**14.94**	94.14	**18.52**
Llama2-13b	90.22	**18.64**	92.12	**20.28**
Llama2-7b	70.77	**21.51**	81.17	**22.16**

2. Consistent preferences across various generators and readers



2. Consistent preferences across various retrievers

Retriever	Generator & Reader	DiffGR
BM25	Llama2-7b	0.5070
Contriever	Llama2-7b	0.5016
Gold	Llama2-7b	0.4656

Why LLMs Prefer Generated Contexts?

1. Confirmation bias is not a key factor

Reader	Generator	ρ_{gen}	ρ_{ret}	ρ_{llm}	DiffGR	DiffGR (ori)
NQ-AIR						
GPT 3.5	GPT 3.5	67.83	12.91	**0.88**	0.68	0.66
GPT 3.5	Llama2-13b	67.22	15.34	**1.20**	0.63	0.58
Llama2-13b	GPT 3.5	62.39	26.76	**1.08**	0.40	0.42
Llama2-13b	Llama2-13b	69.92	18.67	**1.43**	0.58	0.60
TQA-AIR						
GPT 3.5	GPT 3.5	72.05	16.15	**1.55**	0.63	0.61
GPT 3.5	Llama2-13b	70.21	16.38	**2.61**	0.62	0.62
Llama2-13b	GPT 3.5	61.88	26.34	**1.71**	0.40	0.35
Llama2-13b	Llama2-13b	72.55	17.75	**1.96**	0.61	0.58

- Rarely choosing parametric knowledge.
- Still prefer generated contexts even when they are inconsistent with LLMs' parametric knowledge.

2. Semantic completeness matters

Context	Completeness		Similarity		Length
	Sentence	Semantic	Jaccard	BERTScore	
Retrieved	✗	✗	0.1162	0.8552	107.1
Nature	✔	✔	0.1748	0.8730	105.7
S-Trunc.	✔	✗	0.1733	0.8730	105.8
Trunc.	✗	✗	0.1769	0.8736	107.1

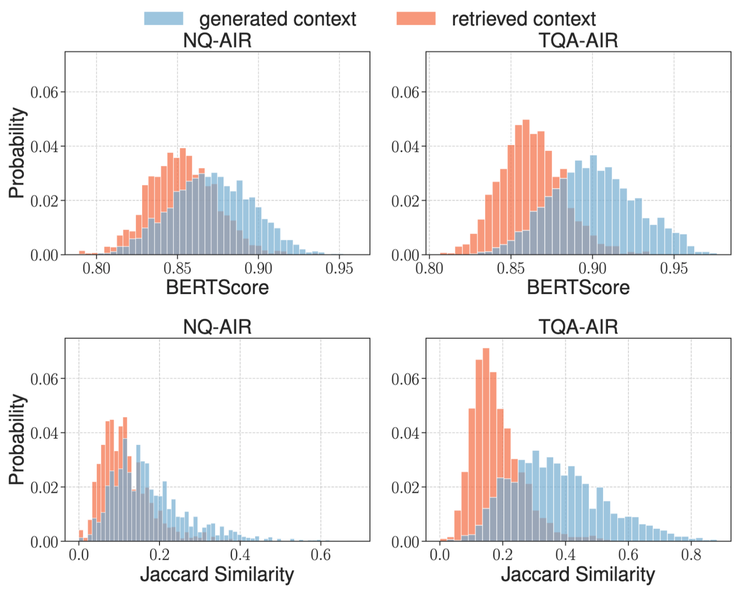
- Simulate incompleteness in generated contexts.
- Controlled length and similarity.

Context Pair	NQ-AIR		TQA-AIR	
	GPT-3.5	Llama2-13b	GPT-3.5	Llama2-13b
Nature vs. Ret.	**0.7519**	**0.6082**	**0.6353**	**0.6207**
S-Trunc. vs. Ret.	0.4864	0.2551	0.5802	0.2779
Trunc. vs. Ret.	0.4792	0.2198	0.5663	0.2787

- Significant increase in bias with enhanced semantic completeness.

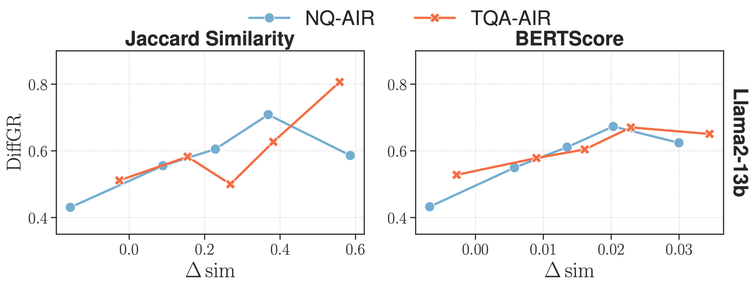
3. Similarity is a significant factor

(1) Similarity Distribution



- Generated contexts exhibit a higher similarity to the questions.

(2) Effect of Similarity



- LLMs exhibit an increased bias on slices with a larger average similarity gap